Run Peng

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EDUCATION

University of Michigan, Ann Arbor - MS in *Computer Science (GPA: 4.0)* *Sep.2022 – Apr. 2024*
University of Michigan, Ann Arbor - BSE in *Computer Science* *Sep.2020 – Apr. 2022*
Shanghai Jiao Tong University - BSE in *Electrical and Computer Engineering* *Sep.2018 – Aug. 2022*

RESEARCH & PROJECT EXPERIENCE

Towards A Holistic Landscape of Situated Theory of Mind in Large Language Models
*Ziqiao Ma, Jacob Sansom, **Run Peng**, Joyce Chai; In submission* *June 2023 – Present*

- Collaboratively designed and executed ten distinct tasks in a 2D grid-world environment, accommodating both single and multi-agent scenarios. Engaged in a comprehensive comparison study, assessing the performance of Large Language Models (LLMs) across seven facets of Theory of Mind.

Robotics Object Tracking with Humans in the Loop
Advisor: Joyce Chai, University of Michigan, with her PhD student Ziqiao (Martin) Ma *Jan 2023 – Present*

- Conducted comprehensive research on the performance of traditional Computer Vision (CV) tasks within real-world robotics contexts, focusing particularly on the impact of human activities leading to alterations in object states.
- Collected and analyzed a multi-object tracking dataset, captured in a residential-like setting using a physical robot.
- Developed various scenario designs for different dataset settings to accommodate diverse objectives, considering factors such as human or robot movements, tasks with different complexities. These scenarios aimed to provide a comprehensive understanding of different interactive environments.

CommonGrid: Building Common Ground through Belief Maintenance in Situated Communication
Advisor: Joyce Chai, University of Michigan, with her PhD student Ziqiao (Martin) Ma *Oct 2022 – Present*

- Constructed a multi-agent interactive simulator for collaborative tasks setting and implemented a user-friendly python-based website platform for human data collection.
- Applied Reinforcement learning for collaborative task completions and encouraged various forms of communications, including natural language dialogues and belief sharing, for better performance. We would like to investigate how maintaining common ground during collaboration helps to optimize the solution to tasks.

Go Beyond Imagination: Maximizing Episodic Reachability with World Models
*Yao Fu, **Run Peng**, Honglak Lee; ICML 2023 Poster* *Dec 2022 – Mar 2023*

- Pioneered a unique method integrating long-term and episodic memory in exploration, further enhancing computational modeling.
- Successfully implemented four benchmark baselines, ranging from procedurally generated sparse-reward 2D environments and continuous control domains, reinforcing its broad applicability and robustness.

Learning Exploration Policies with View-based Intrinsic Rewards
Yijie Guo, Yao Fu*, **Run Peng**, Honglak Lee; NeurIPS 2022 DeepRL Workshop* *Apr 2022 – Aug 2022*

- Performed hypothesis testing, and ablation studies of a novel intrinsic reward design for reinforcement learning; Succeeded in both 2D and 3D environments, and under both task-oriented and task-agnostic settings. Our model significantly outperforms the SOTA.
- Designed most of the visualization, including 2D and 3D environment rendering, agent trace and observed area mapping, and various graphs required in the paper.

Unsupervised Task Clustering with Multi-routed Policy Network
Advisor: Honglak Lee, University of Michigan, with his PhD student Yijie Guo *Oct 2021 – Mar 2022*

- Implemented a multi-routed policy network and applied unsupervised clustering to enable automatic task assignment.
- Investigated AI's behaviors under different dynamics and reward functions; Figured out the generalization quality.

Lightweight of Low light Video Enhancement Algorithm
Advisor: Yinqiang Zheng, University of Tokyo *May 2021 – Dec 2021*

- Improved low-light video quality by designing a novel neural network that places attention on binary binning.

Tradeoff between Network Latency and Migration Cost for Edge Service Migration: A Marriage Problem Strategy via Reinforcement Learning
Advisor: Xudong Wang, Shanghai Jiao Tong University *Feb 2021 – Dec 2021*

- Designed a special state space which smartly stores the data of 5G base stations, edge servers, and autonomous vehicles.
- Created an action transformer in reinforcement learning to optimize service migration strategies in 5G network.

Experimental Evaluation of Deep Reinforcement Learning Algorithms
Advisor: Paul Weng, Shanghai Jiao Tong University *Nov 2019 – Feb 2020*